BSS VS BMS

Kretto

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BSS VS BMS

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Comparison Dimension	Traditional BMS	BSS System (including Mini BSS)
System Definition	Basic battery management system, monitoring voltage, current, and temperature with simple protection functions. Some versions support basic data upload.	Next-generation intelligent battery safety system, integrating big data, AI analysis, multi-physics monitoring, and edge computing, providing proactive prediction, protection, and lifecycle management.
Number of Monitoring Dimensions	3 (voltage, current, temperature).	13 (voltage, current, temperature + pressure + 9 additional electrochemical, mechanical, and behavioral feature signals).
Types of Monitored Signals	Basic electrical signals only, cannot detect internal physical/chemical changes.	Comprehensive coverage of electrical, thermal, mechanical, impedance, and aging indicators, enabling microscopic mechanism-level monitoring.
Big Data Capability	Limited data uploads of current status; insufficient historical data accumulation.	Over 600,000 sets of battery operational data across 6,000 types of battery cells, supporting deep AI learning and optimization.
AI Intelligence Analysis	None; relies on simple threshold-based alarms.	Local edge AI pre-analysis + cloud-based deep learning for dynamic, self-optimizing models.
Warning Accuracy Rate	40%-60%, mostly delayed warnings.	The BSS system delivers an early warning accuracy of over 94%, with an average advance notice of 11.4 hours, while the Mini BSS achieves a 100% early warning accuracy.
Fault Localization Precision	Pack or module level; rough identification.	Precise localization down to individual cells, enabling targeted isolation and repair.

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Comparison Dimension	Traditional BMS	BSS System (including Mini BSS)
Thermal Runaway Prediction Capability	Relies on lagging temperature anomaly detection; high risk of missed alarms.	Real-time detection of pressure changes, impedance shifts, and SOC fluctuation trends; highly accurate and proactive.
Uploaded Data Content	Simple data such as voltage, current, temperature, and SOC estimation.	Full data sets, including original electrical signals, pressure signals, impedance changes, charge/discharge behavior patterns, aging curves, and abnormality detection results.
Edge Computing Capability	None.	Local edge computing filters and extracts valuable data before upload, greatly reducing network and cloud load while improving data quality.
Data Quality for Upload	Raw data with high noise and low information density; difficult for effective analysis.	Pre-processed, noise-filtered, and feature-extracted data with high information density, enabling faster and more accurate analytics.
Alarm System Design	Single-level alarm; simple visual/audible alerts without severity differentiation.	Multi-level alarms (e.g., warning, caution, danger) pushed to user terminals, with actionable maintenance recommendations.
Alarm Precision	Cannot pinpoint faulty individual cells; only detects module/pack-level issues.	Pinpoints the exact faulty cell, providing clear root cause identification and reducing maintenance and replacement costs.

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Comparison Dimension	Traditional BMS	BSS System (including Mini BSS)
Data Processing Method	Centralized cloud processing, slow response, and heavy server load.	Local quick decision-making + cloud deep optimization, enabling real-time, efficient responses.
System Flexibility and Scalability	Fixed function architecture; difficult to upgrade or adapt to different applications.	Modular, software-defined system; flexible expansion and customization across various industries.
Battery Passport and Traceability	None; unable to track battery lifecycle.	Embedded PUF anti-counterfeiting chips to enable production-to-recycling full lifecycle traceability and authentication.
Miniaturization Application (Mini BSS)	Not suitable for small battery devices.	Mini BSS adopts ultra-thin flexible sensor technology, perfectly fitting lightweight EVs, power tools, portable storage, and consumer electronics.
Maintenance and Upgrade	Manual inspections needed; long cycles; delayed problem detection.	Remote real-time monitoring, online upgrades, and proactive maintenance alerts, significantly lowering O&M costs.
Compliance with Policies	Partially compliant with basic safety regulations; weak adaptability to future carbon neutrality and recycling requirements.	Fully aligned with global standards for safety monitoring, carbon neutrality, battery recycling, and lifecycle traceability.
Market Growth Prospects	Aging technology, slowing growth, and fierce competition.	Emerging leader in the smart energy management field; high technical barriers, broad market potential.

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Key Summary

Traditional BMS Warning:

➔ Rough, delayed alarm at module or pack level,
➔ No ability to precisely identify faulty cells,
➔ Cannot guide timely maintenance actions.

BSS (including Mini BSS):

➔ Multi-level alarms (warning/caution/danger),
➔ Accurate fault pinpointing down to individual cells,
➔ Provides users with clear maintenance recommendations,
➔ Local edge computing filters noisy data,
➔ Uploads only valuable features,
➔ Greatly reduces network and cloud burden, while improving overall data efficiency.
➔ **BSS is not just about "reporting failures after they occur," but "preventing failures before they happen."**
➔ **BSS is not simply "raising an alarm," but "providing precise diagnosis and action plans."**
➔ **BSS represents the intelligent safety system that the energy industry truly needs for the future.**